Point vs interval estimation + estimator characteristics

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Asked to distinguish one estimate from a range.
- Need to judge an estimator by repeated-sampling behaviour.
- Parameter is unknown; statistic estimates it.

2 DECISION / METHOD

1. Match parameter to statistic: μ <- X̄; π <- p.
2. Point estimate = one statistic value.
3. Interval estimate = plausible range plus confidence level.
4. Judge unbiasedness, consistency and relative efficiency.

3 FORMULA

```
unbiased: E(estimator)=parameter
consistent: error shrinks as n grows
more efficient: smaller variance
among unbiased estimators
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: parameter/statistic pairings and standard errors.
Still know: definitions of point, interval, unbiasedness, consistency and efficiency.

5 EXCEL ROUTE

- Mean point estimate: `=AVERAGE(range)`
- n: `=COUNT(range)`
- Proportion point estimate: successes/count.

6 TINY WORKED EXAMPLE

A sample gives X̄=42.6. That is a point estimate of μ. A 95% CI such as (40.1,45.1) adds precision information.

7 TOP EXAM TRAPS

- Saying the parameter changes from sample to sample.
- Unbiased means 'always exactly correct'.
- Calling a confidence level a probability that this fixed interval is correct.

8 SAY ALOUD

The statistic is my sample-based point estimate; the confidence interval shows its uncertainty about the fixed parameter.

9 EXIT CHECK

Which estimator characteristic prefers smaller variance when both estimators are unbiased?

ONE-GLANCE VISUAL ROUTE

